

01/11/2025 Morning

The topics that we discuss today: Velocity rise, impulsive velocity rise, gravity, ^{Drag} Example,

We have acceleration Eqⁿ: $\frac{du}{dt} = \frac{m u_{eq}}{M} - g \cos \theta - \frac{D}{M}$

$\dot{m} = -\frac{dM}{dt} \rightarrow$ put in above Eqⁿ: $\left[\frac{du}{dt} = \frac{u_{eq}}{M} \frac{dM}{dt} - g \cos \theta - \frac{D}{M} \right]$

$$\rightarrow du = -u_{eq} \frac{dM}{M} - g \cos \theta dt - \frac{D}{M} dt$$

Now, integrate this Eqⁿ:

$$\int_{u=u_0}^{u=u_b} du = - \int_{M_0}^{M_b} u_{eq} \frac{dM}{M} - \int_{t=0}^{t=t_b} g \cos \theta dt - \int_{t=0}^{t=t_b} \frac{D}{M} dt$$

$$u_b - u_0 = \Delta u_b = u_{eq} \ln\left(\frac{M_0}{M_b}\right) - \overline{g \cos \theta} t_b - \left(\frac{\overline{D}}{M}\right) t_b$$

$$\overline{g \cos \theta} = \frac{\int_{t=0}^{t=t_b} g \cos \theta dt}{t_b} \quad \& \quad \frac{\overline{D}}{M} = \frac{\int_{t=0}^{t=t_b} \frac{D}{M} dt}{t_b}$$

Impulsive velocity rise ($\Delta u_{b,imp}$)

$\Delta u_{b,imp}$ corresponds to $t_b = 0$

$$\Delta u_{b,impulsive} = u_{eq} \ln\left(\frac{M_0}{M_b}\right)$$

$$\Rightarrow \Delta u_b = \Delta u_{b,imp} - \overline{g \cos \theta} t_b - \left(\frac{\overline{D}}{M}\right) t_b$$

(P-1)

gravity

$$\frac{g}{g_e} = \left(\frac{R_e}{R_e + h} \right)^2 \quad \boxed{R_e = 6378 \text{ km}}$$

We could safely assume $g = \text{cte}$ in this course (Not always)

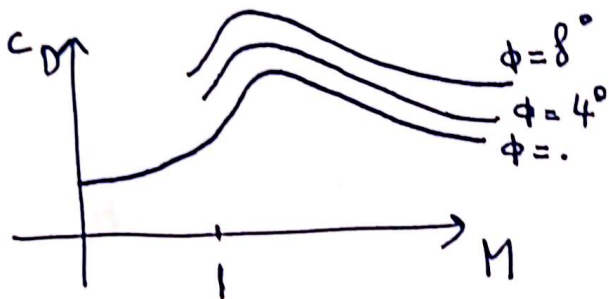
Drag

$$D = c_D \frac{1}{2} \rho u^2 A_f$$

$$\rho = a \exp(-bh^{1.15}) \quad \text{where } a = 1.2, b = 2.9 \times 10^{-5}$$

$$h \text{ in (m)}, \rho \text{ in } \frac{\text{kg}}{\text{m}^3}$$

c_D is a function of M & Reynolds



see Fig 10.4 of textbook

Example: A rocket with following data:

$$\dot{m} = 87410 \frac{\text{kg}}{\text{s}}, u_e = 2600 \frac{\text{m}}{\text{s}}, p_e = 71 \text{ kPa}, A_e = 11.35 \text{ m}^2$$

$$M_o = 1,720,000 \text{ kg}, M_b = 680,000 \text{ kg}, p_a = 101 \text{ kPa at SL}$$

$$\theta = 0, g = g_e, D = 0$$

$$\text{Find: } \tau_{\text{SL}}, u_{\text{eq,SL}}, I_{\text{sp,SL}}, u_{\text{eq,vac}}, I_{\text{sp,vac}}, a_o, a_b$$

$$a_b^+, \Delta u_b$$

$$\tau = \dot{m} u_e + (p_e - p_a) A_e$$

$$\tau_{SL} = 8740 \times 2600 + \underbrace{(71 - 101) \times 10^3}_{\text{is negative! over-expanded nozzle!}} \times 11.35 = 2.24 \times 10^7 \text{ N}$$

$$u_{eq, SL} = \frac{\tau_{SL}}{\dot{m}} = \frac{2.24 \times 10^7}{8740} = \underbrace{2563 \frac{\text{m}}{\text{s}}}_{u_{eq, SL}} < \underbrace{2600 \frac{\text{m}}{\text{s}}}_{u_e}$$

$$I_{sp, SL} = \frac{u_{eq, SL}}{g_e} = \frac{2563}{9.81} = 261 \text{ s}$$

$p_a = 0$ in vacuum

$$\tau_{vac} = \dot{m} u_e + p_e A_e = 8740 \times 2600 + 71 \times 10^3 \times 11.35 = 2.35 \times 10^7 \text{ N}$$

$$u_{eq, vac} = \frac{\tau_{vac}}{\dot{m}} = \frac{2.35 \times 10^7}{8740} = 2689 \frac{\text{m}}{\text{s}} > 2600 \frac{\text{m}}{\text{s}}$$

$u_{eq, vac} > u_e$

$$I_{sp, vac} = \frac{u_{eq, vac}}{g_e} = \frac{2689}{9.81} = 274 \text{ s} > I_{sp, SL}$$

$$a_o = \frac{\dot{m} u_{eq, SL}}{M_o} - g_e = \frac{8740 \times 2563}{1720000} - 9.81 = 3.21 \frac{\text{m}}{\text{s}^2} = 0.33 g_e$$

$$a_{b-} = \frac{\dot{m} u_{eq, SL}}{M_{b-}} - \frac{g_e}{\cos \theta_b} - \frac{\dot{D}_b}{M_b} = \frac{8740 \times 2563}{680000} - 9.81 = 23.1 \frac{\text{m}}{\text{s}^2} = 2.36 g_e$$

$$a_{b+} = \frac{\dot{m} u_{eq, SL}}{M_{b+}} - \frac{g_e}{\cos \theta_b} - \frac{\dot{D}_b}{M_b} = -g_e = -9.81 \frac{\text{m}}{\text{s}^2}$$

P-3

$$\text{Burn time } t_b = \frac{M_o - M_b}{\dot{m}} = \frac{1,720,000 - 680,000}{8740} = 119 \text{ s}$$

$$\Delta u_b = u_{eq} \ln\left(\frac{M_o}{M_b}\right) - \int_0^{t_b} \underbrace{g \cos \theta}_{g_e} dt - \int_0^{t_b} \frac{\dot{D}}{M} dt$$

$$\Rightarrow \Delta u_b = u_{eq} \ln\left(\frac{M_o}{M_b}\right) - g_e t_b = 1.21 \frac{\text{km}}{\text{s}}$$
